Instructions

- Submit the assignments in time via online form here.
- The solutions MUST be done in LaTeXand handed as pdf. No handwritten assignments will be accepted.
- For python based problems, provide the codes as well as any generated figure. Seek your instructor's or peers' help if you need it. Responses during the viva that amount to: '...that's what the AI gave me', would result in a 0.
- Equations must be numbered and properly formatted.
- Expect some simple questions from the readings/viewings in the exam as well.
- Your final marks in the assignment will be contingent on the viva.

Gauge Transformations

1. Griffiths 10.7

Retarded Potentials

2. Griffiths 10.9, 10.10

Point Charges

3. Griffiths 10.13

Python

4. Matplotlib can be used to make animations. Create a simple animation of a red particle moving along an elliptical path. The parameters of the ellipse are up to you.

   In the next assignment you will be asked to animate Figure 10.7 of Griffiths.

Reading/Writing/Viewing

5. Watch:

   (a) Solving differential equations with python/numpy.

   (b) Laplace's Equation with Arbitrary Boundary Conditions in python

   Your second project will revolve around modeling Laplace's equation in Python.